i2X Forum for the Implementation of Reliability Standards for Transmission (FIRST) Summary Notes

Interconnection Innovation e-Xchange (i2X)

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Working Draft

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Disclaimers

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Executive Summary

The DOE i2X Forum for the Implementation of Reliability Standards for Transmission (FIRST) established an open industry forum to facilitate discussion, brainstorming, and information sharing regarding the supports the adoption of new standards and test procedures for inverter-based resources (IBRs)[1] connecting to the transmission and subtransmission electric system. Building on the 2024 *Transmission Interconnection Roadmap*, i2X FIRST convenes industry stakeholders to share practices on standard implementation of IEEE 2800-2022 and the upcoming IEEE P2800.2 recommended practices through webinars and workshops. Topics include IBR ride-through, modeling, monitoring, frequency and voltage support, and evolving technologies like grid forming inverters. Discussions also align with FERC directive and ongoing NERC standards revisions.

Season 1 of the i2X FIRST initiative provided a foundational exploration of IEEE 2800-2022, its core requirements, and early industry experiences with implementation. Season 2 continued these efforts, with a focus on IBR plant design evaluations, change management during the interconnection process, IBR plant commissioning best practices, and emerging standards developments and technology adoption. Meeting recordings, presentations, and a full synopsis and brief recap summary from Season 1 and Season 2 are available here and here, respectively. Sign up for Season 3 i2X FIRST meetings here.

[1] Such as solar photovoltaic (PV), wind, and battery energy storage systems (BESS), and hybrid plants comprised of these technologies.

List of Technical Acronyms

AC	Alternating Current
ACE	Area Control Error
AGIR	Authority Governing Interconnection Requirements
AVR	Automatic Voltage Regulator
BOP	Balance of Plant
BPS	Bulk Power System
CHIL	Control Hardware in the Loop
COD	Commercial Operation Date
CPR	Duke Capability and Performance Report
DC	Direct Current
DER	Distributed Energy Resource
DFR	Digital Fault Recorder
DFT	Discrete Fourier Transform
DLL	Dynamic Link Library
EMT	Electromagnetic Transient
ESR	Energy Storage Resource
FAT	Factory Acceptance Testing
FFR	Fast Frequency Response
FRT	Frequency Ride-Through
GFL	Grid Following
GFM	Grid Forming
GIA	Generator Interconnection Agreement
GIP	Generator Interconnection Procedure
GO	Generator Owner
GOP	Generator Operator
HIL	Hardware in the Loop
HVDC	High Voltage Direct Current
IBR	Inverter-Based Resource
IFRO	Interconnection Frequency Response Obligation
IPID	IBR Plant Information Database
IRR	Intermittent Renewable Resource
ISO	Independent System Operator
LSL	Low Sustained Limit
OEM	Original Equipment Manufacturer
OLTC	On-Load Tap Changer
PC	Planning Coordinator
PCS	Power Conversion System
PDT	Phasor Domain Transient
PFR	Primary Frequency Response
PLL	Phase Lock Loop

PMU	Phasor Measurement Unit
POC	Point of Connection
POI	Point of Interconnection
POM	Point of Measurement
PPC	Power Plant Controller
RFI	Request for Information
ROCOF	Rate of Change of Frequency
RMS	Root Mean Square
ROI	Return on Investment
RPA	Reference Point of Applicability
RTO	Regional Transmission Organization
RVC	Rapid Voltage Change
SAT	Site Acceptance Testing
SCADA	Supervisory Control and Data Acquisition
SFTP	Secure File Transfer Protocol
SSCI	Subsynchronous Control Interactions
SCR	Short Circuit Ratio
SIL	Software in the Loop
SMIB	Single Machine Infinite Bus
SSR	Subsynchronous Resonance
TOV	Transient Overvoltage
TP	Transmission Planner
TS	Transmission System
TSO	Transmission System Operator
UDM	User-Defined Model
UFLS	Underfrequency Load Shedding
VRT	Voltage Ride-Through
VSC	Voltage Source Converter
WSCR	Weighted Short Circuit Ratio
WTG	Wind Turbine Generator

* Generally not including names of organizations, institutions, or initiatives.

June 4, 2026 Virtual Meeting

Deep Dive PRC-029 Implementation Process Updates (~XYZ attendees)

Presentation recording and slides are available to download here. Figure 1 shows the makeup of meeting attendees by industry sector:

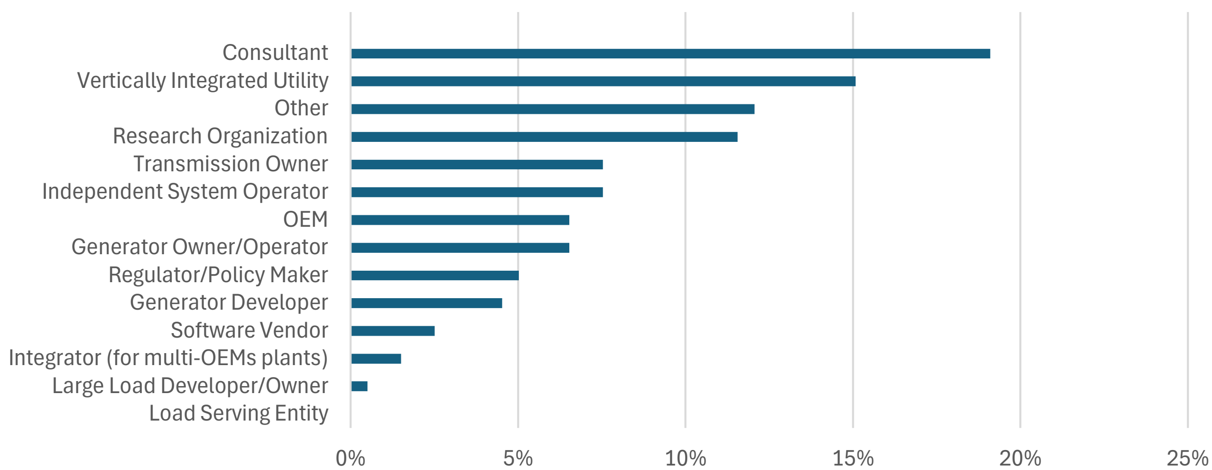

Figure 1: Meeting attendees by industry sector

This first meeting of Season 3 focused on implementation of NERC PRC-029-1, developer and generator owner perspectives, relation to IEEE 2800.2 tools, draft NERC Implementation Guidance, and revisions to PRC-029-2 related to FERC Order 909. Presentations included the following:

Eugen Starschich, Siemens Energy

Eugen provided an update on the Project 2025-05 Ride-Through Revisions and the modifications to PRC-029-1 to address FERC directives in Order 909. FERC Order No. 909 approved PRC-029-1 and its implementation plan. PRC-029-1 becomes effective in the United States on October 1, 2026; FERC directed NERC to submit a responsive modification to the standard within twelve months of the rule's effective date. The schedule in Figure 2 shows the anticipated actions for the remainder of 2026.

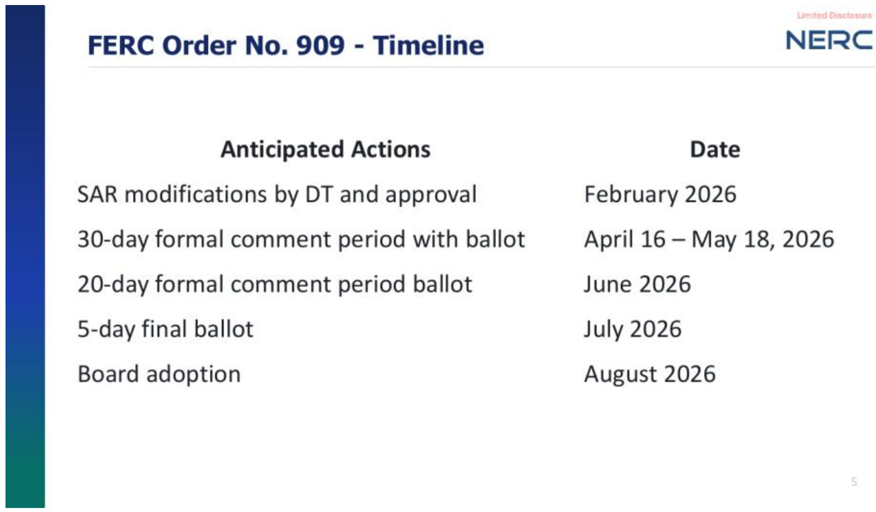

Figure 2: Project 2025-05 development timeline.

The FERC directive identified three concerns the modification must address:

1. **Clarity on documentation obligations supporting hardware-limitation exemptions for legacy IBRs.** The drafting team chose not to publish a non-exhaustive list of acceptable documents, concluding that such a list would only generate additional questions about anything omitted. Instead, PRC-029-2 establishes documentation categories that provide technical evidence of a hardware limitation: equipment vendor documentation of the limitation; written engineering analysis; test results; and operational data experience supported by an engineering analysis. A generator owner claiming an exemption must justify this using one or more of these categories. The team also addressed ambiguity regarding replacement of existing hardware causing a limitation., adding a clarifying note that the hardware causing the limitation shall be understood as an integrated system of components, where integrated system means a combination of components such that replacing any subset of them would not allow the combination to remove the previously documented limitation. Some examples included:
    - **Example 1:** A wind turbine converter is the limiting hardware, and converters are planned for replacement. No compliant replacement converter is commercially available or technically feasible (whether because none exists, because a compliant unit would push other turbine components outside their design envelope, or because it simply does not fit dimensionally). In that case, the exemption applies to the entire integrated set that would need replacement – i.e., the entire wind turbine and its subcomponents, from converter and generator through nacelle, hub, blades, pitch and yaw systems, tower, and foundation.
    - **Example 2:** Converters on roughly 10% of a plant's turbines are replaced due to equipment failures with a commercially available converter capable of meeting

the requirements; even though that subset is now capable, the exemption remains at the plant level.

2. **Limited exemptions for IBR facilities under development but not yet in service on the standard's effective date.** Facilities designed and procured under old rules that will not be in service until after October 1, 2026, could run the risk of long lead-time equipment not being compliant with the new rule. PRC-029-2 extends hardware-limitation exemption eligibility to such new IBRs if the project meets the criteria:
    - An executed interconnection agreement by the effective date of PRC-029-1, and
    - An executed equipment procurement contract for the hardware causing the limitation prior to the first day of the first month following the FERC effective date of PRC-029-1.

    The drafting team emphasizes that these dates align with FERC's directive to keep allowable exemptions limited in scope. To remove ambiguity about what "in service" means, a new footnote ties the term to the FERC pro forma Large Generator Interconnection Agreement, under which the In-Service Date is the date the interconnection customer reasonably expects to be ready to begin using the transmission provider's interconnection facilities to obtain ***back-feed power***.

3. **Targeted, limited exemptions for HVDC-connected IBRs.** This issue arises from architectures where an HVDC link connects a remote or islanded network such as offshore wind (see Figure 3). Because the HVDC link decouples the islanded network from transmission-system faults, the energy that the islanded generators continue to produce during a fault has nowhere to go. It is dissipated as heat in chopper resistors, and this enables reliable power recovery after faults. This could be construed as in conflict with PRC-029-1 Attachment 1, items 7 through 9, which make ride-through durations cumulative over any ten-second window and prohibit tripping for four or fewer voltage deviations within such a window. There are technical limitations to chopper circuit capabilities that needed to be addressed. The proposed solution includes a new item 12 in PRC-029-2 Attachment 1 which exempts a VSC-HVDC-connected IBR from items 7, 8, and 9 when it includes an AC or DC chopper and reaches its designed hardware thermal energy absorption limit. Some limiting conditions are included for implementation clarity.

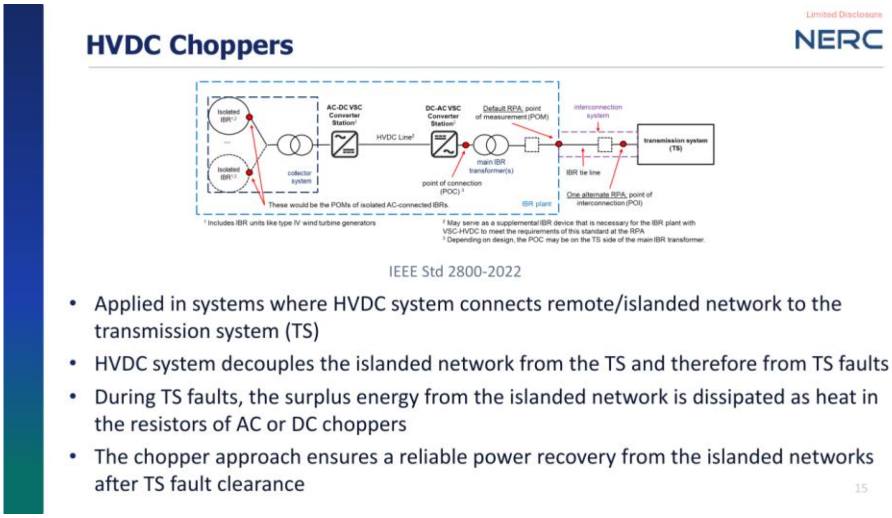

Figure 3: HVDC chopper architecture and discussion points.

<u>Jens Boemer, EPRI</u>

Jens presented on leveraging IEEE 2800 family of standards for alignment and compliance with NERC PRC-029-1. Most PRC-029-1 requirements map to IEEE 2800-2022 requirements and IEEE 2800.2-2026 test procedures, within some known differences between the standards.

IEEE 2800.2-2026 is the newly published Recommended Practice that harmonizes test and verification procedures for conformity assessment of IBR plants. The set of recommended practices set a foundation enabling a paradigm shift toward conformity assessment, which is a structured, staged framework (see Figure 4) rather than ad-hoc compliance demonstrations.

That framework runs from type tests of IBR units, through IBR unit model validation, IBR plant model development, IBR plant design evaluation by simulation, as-built installation evaluation once construction is complete, and onward to commissioning tests, post-commissioning model validation, post-commissioning monitoring during grid events, and periodic tests and verifications across the IBR plant's life. IBR plant design evaluation and as-built installation evaluation are two key steps involved with PRC-029-1 compliance.

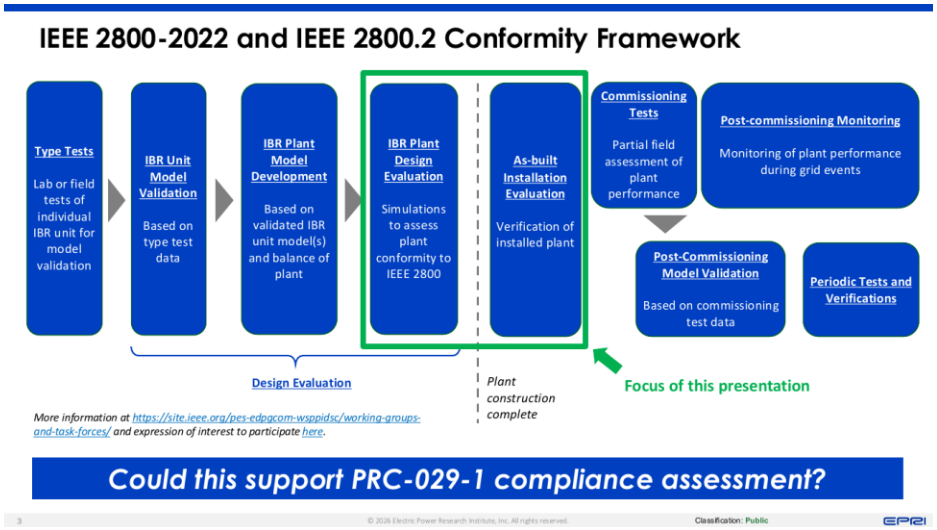

Figure 4: IEEE 2800/2800.2 conformity assessment framework.

PRC-029-1 establishes ride-through requirements similar to IEEE 2800-2022 but leaves out some important technical details on performance (see Figure 5).

For PRC-029-1 Requirement R1, the IEEE 2800.2 balanced and unbalanced low-voltage and balanced high-voltage disturbance ride-through test tables provide direct verification procedures. Requirement R1 has more nuances exceptions. Tripping to clear a fault and tripping when voltage exceeds a documented hardware limit have no analog in IEEE 2800.2 tests. Exception for instantaneous positive-sequence phase angle shifts of 25 degrees at the high side of the main power transformer does map to IEEE 2800.2's phase-angle-change ride-through tests and is supported. Regarding the V/Hz exception, IEEE 2800 recognizes ride-through within "capability limits" but never specifies a definitive V/Hz limit.

IEEE 2800.2 has a defined set of tests to demonstrate conformity with the voluntary standard; on the other hand, PRC-029-1 Measures offer a non-exhaustive list of examples such as dynamic simulations, studies, and protection and control settings evaluations to demonstrate compliance (similar to Measures in other NERC Standards). On the operational side of the Measures, retaining disturbance monitoring evidence to show IBR plants rode through actual events (and may have exceptions apply) helps ensure evidence to those events.

Some differences in IBR plant ride-through behavior and how that is assessed between IEEE 2800.2 and PRC-029-1 were discussed, particularly the requirements being applicable at the IBR unit level versus IBR plant level.

Requirement R3 frequency ride-through is generally conducted through OEM and plant design documentation rather than simulation tests.

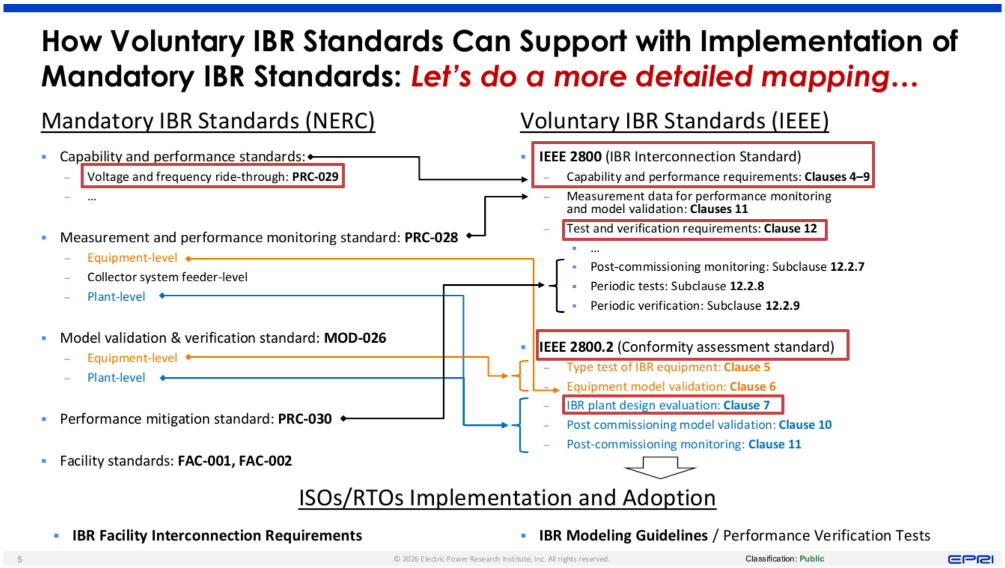

Figure 5: EPRI mapping of mandatory NERC IBR standards (left) to voluntary IEEE 2800 and 2800.2 clauses (right).

EPRI described their IBR Conformity Assessment (IBR-CA) Module release, which is an independent software tool that determines whether a given IBR plant model or measured IBR plant performance conforms to a specific standard or grid code, currently focused on IEEE 2800-2022 and 2800.2-2026 tests and criteria. EPRI also discussed a central certificate register, pointing to Germany's ZEREZ database, as well as outlined related offerings including the Application of IBR Standards Collaborative Forum.

Tim Taylor, SEIA; Ryan Quint, Elevate Energy Consulting

Tim Taylor (SEIA) and Ryan Quint (Elevate) walked the industry through a draft Implementation Guidance (IG) document that SEIA and its members developed for Requirements R1 through R3 of PRC-029-1. The presentation provided practitioner perspectives on demonstrating compliance with the IBR design aspects of PRC-029-1 by the October 1, 2026 deadline for bulk electric system (BES) IBRs and January 1, 2027 for Category 2 IBRs.

NERC Compliance Implementation Guidance is developed by industry, for industry, and is vetted and submitted through pre-qualified organizations; SEIA recently attained this qualification in January 2026 and this was one of the first deliverables to support this initiative. An IG document does not create new requirements or define the only acceptable approach; it solely illustrates examples or approaches that registered entities could use to implement a Reliability Standard. If the ERO Enterprise endorses the guidance, auditors give the described approach deference during Compliance Monitoring and Enforcement Program (CMEP) activities. At the time of presentation, the document was still in draft status under ERO Enterprise review; however, it provides clear and comprehensive technical considerations and methodologies for industry awareness and understanding.

The IG includes a six-step evaluation approach, illustrated in Figure 6. The protection and control setting verification and ride-through simulations are flagged as the key assessment steps and the remainder of tasks are supporting activities. Wrapping around the entire process is coordination with original equipment manufacturers (OEMs), reflecting the reality that inverter and plant controller vendors hold much of the information a generator owner needs. Third parties such as EPC firms, protection engineers, field personnel, SCADA engineers, and modeling experts may also be required.

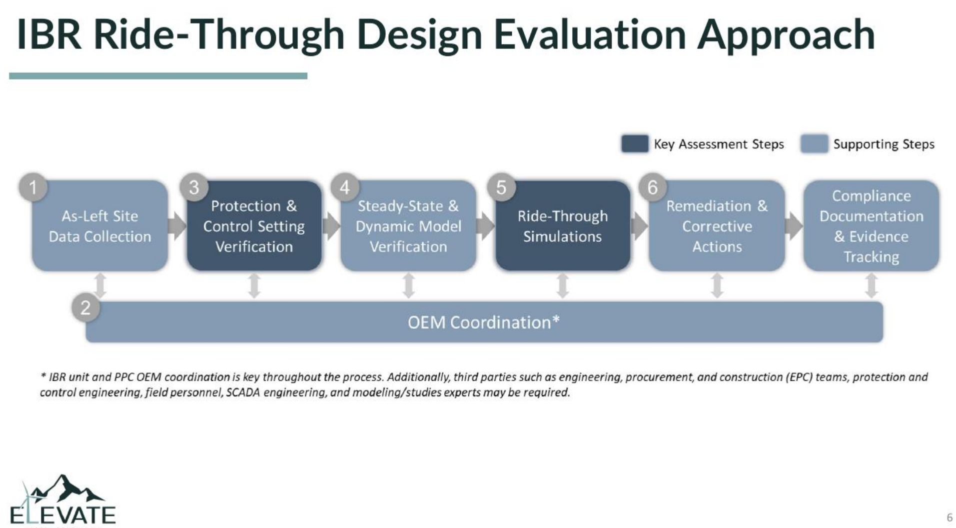

Figure 6: SEIA IG six-step IBR ride-through design evaluation approach.

The framework includes the following steps:

1. **Step 1: As-left site data collection.** General site information, generator interconnection agreement, back-feed date, as-left balance-of-plant (BOP) relay settings, as-left inverter protection and control settings, single-line diagrams, unit manuals, aggregate dynamic models, rate-of-change-of-frequency (ROCOF) protection information, consecutive voltage disturbance ride-through capability, and plant controller settings; supplementary information such as control narratives, capability curves, PRC-019/PRC-024 reports, MOD test reports, transformer factory acceptance tests, etc.
2. **Step 2: OEM coordination.** Early engagement with OEMs is strongly encouraged, and sharing a clear checklist of necessary information can help streamline data collection and review. This includes IBR unit ride-through capability, as-left settings, validated models, phase jump/ROCOF/anti-islanding protections, consecutive disturbance capability, filtering and measurement windows, etc.
3. **Step 3: Protection and control verification.** For IBR units, cover voltage and frequency ride-through settings, voltage and frequency measurement filtering for protection and control, OEM-stated maximum equipment capabilities, ride-through mode thresholds,

momentary cessation and current blocking, current priority during ride-through, fault current injection settings, etc. For PPCs, protection and control settings, control loops, deadbands, droops, and ramp rates. At the BOP relay level, relay settings on collector feeders, medium- and high-voltage buses, tie lines, reactive devices, and V/Hz relaying at the main power transformer.

4. **Step 4: Model verification.** Ensuring the model is reflective of the actual site, and as needed the post-modification state, is important yet extremely challenging, particularly in gathering necessary information and models from OEMs of a future software upgrade product. Careful consideration and workarounds may be needed, but generally the model should be used where possible, with careful engineering consideration.
5. **Step 5: Simulation tests.** Simulations across an array of base cases and simulation conditions test the ride-through performance and IBR plant behavior thoroughly and can be used to corroborate desktop evaluations. Situations do arise where simulations may show ride-through whereas OEM documentation explicitly states a limitation. This is to be expected due to limitations in the dynamic models in simulating some key ride-through limitations such as blade aerodynamics, DC-side limitations, etc.
6. **Step 6: Remediation and corrective actions.** Identifying software, protection, control, configuration, and BOP modifications need to be implemented by the compliance deadlines, and may require iterating the ride-through design validation through multiple rounds of testing. This also involves extensive coordination with IBR plant personnel, OEMs, protection engineers, and compliance teams. The complexity and downstream ramifications of modifications to the site should not be understated, particularly when protection and control settings, and accompanying models, must be modified. Some regions like ERCOT have a strict site modification process that must be accounted for as well in terms of implementation timelines.

Katie Iversen, AES

Katie presented on actual experience of a large renewable fleet owner preparing for and conducting PRC-029 evaluations. Multiple grid disturbances revealed that IBR behavior during disturbances did not support the grid (e.g., tripping or momentary cessation). Industry response accumulated with NERC Alerts, IEEE 2800, PRC-024-3 revisions, ERCOT NOGRR245, and other industry efforts. Finally, PRC-029-1 was developed and approved. Three key points were made:

1. NERC standards exist to support reliability, which is in an asset owner's interest
2. PRC-029 may not be the only ride-through requirement applicable to a given asset; contradictory or additive requirements in interconnection agreements, power purchase agreements, IEEE 2800 adoption by transmission providers, and ERCOT NOGRR245 can each impose overlapping or stricter obligations.

3. Design reports, settings files, and related material are stored and provided for enforcement activities, and this evidence should be produced and retained accordingly.

OEMs publish maximum ride-through settings for their equipment, but maximum capabilities and recommended settings are two different concepts, and equipment frequently ships with the latter. Maximum setting recommendations have changed over time, and maximum settings are not always what gets selected as "recommended" or "default" during commissioning. This introduces compliance exposure and challenges with execution; detailed attachment items in the standard are often overlooked by OEMs and getting answers to these necessary questions has been challenging.

Katie offered parallel to-do lists. OEMs should publish documentation and information pertaining to the requirements set forth, which can quickly and effectively be distributed to customers and serve as a "one stop shop" for requests. OEMs can also maintain revision histories that explain changes and explicitly flag any that introduce lesser performance; changes to documentation are extremely painful for asset owners downstream during evaluations. For developers and generator owners, previous/existing settings are unlikely to comply. Existing IBR unit and BOP relay settings often need to change, which has downstream consequences on protection coordination, asset management, model updates, MOD testing, EPC agreements, and many other factors.

The key is to internalize major changes from the PRC-024 era, get OEM capability documentation in hand, complete assessments and documentation, and prepare for the hardware-based exemptions in a timely manner.

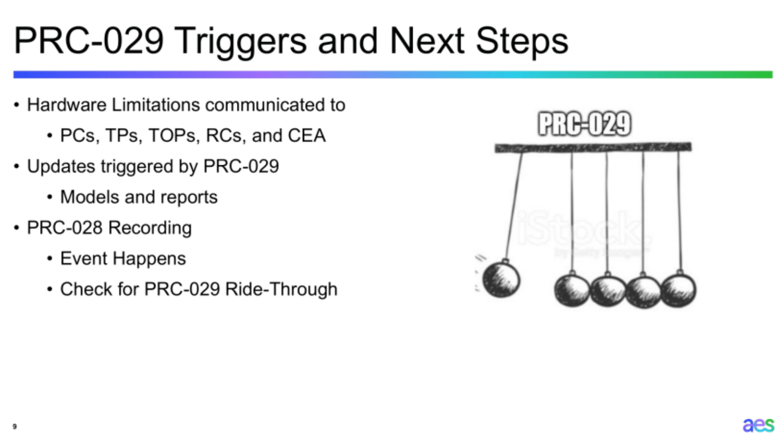

Figure 7: PRC-029 triggers and downstream impacts.

Q&A and Interactive Group Discussion

Why not use "by the effective date of PRC-029-1" for both the executed interconnection agreement and the executed equipment procurement contract? Why add the "first day of the first month" language to the latter?

The NERC compliance team preferred not to use the mid-month date of August 28 as a reference and wanted the beginning of the month as the reference date instead. The "first day of the first month following the FERC effective date" wording was selected for that reason.

A central database of each OEM's equipment capabilities and limitations would be valuable, particularly for PRC-029 R4 documentation and evidence. Is anything like that available?

The panel agreed the idea has merit and pointed to Germany's ZEREZ central certificate register (www.zerez.net) as a potential "proxy" in the absence of a comparable North American database.

Regarding PRC-029 R4 documentation, can the drafting team provide examples of what engineering analysis and test results may look like in the absence of vendor documentation?

The drafting team will explore adding examples to the standard's Technical Rationale, though the tight development schedule may not allow it. Attendees suggested that a future Implementation Guidance document for R4 could also fill this need.

Is there a distinction between hardware and firmware for exemption purposes?

FERC's directive allows exemptions only for hardware limitations, so software and firmware limitations do not qualify. The drafting team agrees firmware limitations are also critical and not easily mitigated but addressing them would require a new Standard Authorization Request.

Is PRC-029-2 expected to be approved before PRC-029-1 compliance must be demonstrated or exemptions sought? How will the two versions align?

The drafting team's target is for PRC-029-2 to become effective on the same date as PRC-029-1, so that the revised exemption provisions are valid from the outset.

Regarding Example #2 (partial converter replacement), wouldn't that potentially trigger FAC-002 qualified change criteria, requiring notification of the PC/TP and a re-study?

Yes, a re-study may be required. However, the exemption can still be applied if the study shows the ride-through requirement cannot be fulfilled.

How do we handle PRC-029-2 compliance for IBRs with legacy inverters or wind turbines whose OEMs are no longer in business?

The hardware limitation exemption can be used by demonstrating the requirements cannot be met. Where vendor documentation is unavailable for such units, a written engineering analysis can serve as the supporting evidence.

Can the German equipment certification database be used for the same equipment in the U.S. to assess PRC-029 compliance?

EPRI's analysis found the technical requirements are "close enough" that German certifications can manage a large part of the conformity risk relative to IEEE 2800 and, by extension, PRC-029. Whether German certificates can legally be relied upon for compliance is a question for attorneys.

For IBRs connecting to 525 kV systems, the upper voltage thresholds can exceed the 550 kV equipment rating. Is there guidance on how to handle this, especially since future projects likely can't rely on exemptions?

The first step is understanding which specific equipment is experiencing the limitation and why. If this proves to be a widespread technology limitation, a new SAR can be submitted to NERC to update PRC-029 accordingly — and anyone may submit a SAR.

OEM engagement is the most critical piece of the IBR ride-through design evaluation and model quality testing. Any suggestions for making this step smoother for the industry?

If all stakeholders, including OEMs, follow the recommended practices in IEEE 2800.2-2026, coordination should become smoother over time. Industry members who find errors or areas for improvement are encouraged to engage through the IEEE PES working groups.

What issues are you seeing in power plant controllers (PPCs) with regard to PRC-029?

PPCs play a meaningful role in compliance assessments, but potential issues often go unnoticed because PPCs are not accurately represented in the models. More rigor is needed to true up models and settings at commissioning; a draft NERC Reliability Guideline on commissioning (NERC IRPS work item #24) is in development and will provide detailed guidance.

Is having this Implementation Guidance endorsed by NERC likely to encourage OEMs to provide standardized information packages for each inverter make and model?

That outcome would be highly desirable. There is no reason each GO or developer should approach the OEM project-by-project for the same make-and-model information — a shared database of equipment capabilities would reduce burden for everyone and improve consistency.

If the model rides through per PRC-029, how do you know what the model shows is correct? What would make you dig deeper and ask OEMs for additional details, such as ROCOF or phase jump relay settings?

IEEE 2800.2-2026 requires thorough validation and verification of the models. Standard library models cannot represent ROCOF or phase angle protections, which is precisely why those settings must be requested from the OEM and verified separately.

We've run into an issue where the model does not show recovery within 1 second of an LVRT event. Is anyone else seeing this, and are there potential fixes?

This is a known issue in the PDT domain, commonly caused by incorrectly modeled PPC "freeze" settings or REEC voltage-dip (Vdip) parameters. Recommended checks include the Vup/Vdip thresholds, deadbands, Vref, Thld, and dpmax/dpmin ramp limits in the REEC model, as well as confirming any OLTC is modeled and enabled properly; EPRI also offered to review specific cases directly.

Key Themes

1. Compliance deadlines are driving urgency across the industry. Generator owners are actively conducting IBR plant ride-through design evaluations, engaging with OEMs, and determining existing and possible modified settings to comply with the standard or seek hardware-based exemptions.
2. OEM coordination is the critical bottleneck, and nearly every presenter and Q&A exchange pointed to OEM documentation, as-left settings information, dynamic models, etc., as the hardest inputs to obtain. There was strong discussion around standardized information packages or a central equipment capability database for these types of evaluations.
3. Voluntary standards and shared guidance can streamline mandatory compliance. IEEE 2800.2-2026 test procedures, EPR IBR-CA tool, and SEIA's draft Implementation Guidance are all aids to meeting PRC-029-1 obligations.
4. Models and settings must reflect field reality: Existing settings are unlikely to comply, standard library models cannot capture key protections such as ROCOF and phase jump, PPC behavior is often misrepresented in simulation models, etc. Protection and control verification, model validation, and site documentation rigor are essential elements to a full ride-through design evaluations (in addition to simulations, where those are possible).

For more information, visit: energy.gov/i2X

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